
Pregunta

a) (1,25 punts) Calcula les integrals $\int (3x^2 + 4x - 2) dx$ i $\int (x - 1)^3 dx$.

Solució. $\int (3x^2 + 4x - 2) dx = x^3 + 2x^2 - 2x + k$ i $\int (x - 1)^3 dx = \frac{(x - 1)^4}{4} + k$.

b) (1,25 punts) Calcula $\int \left(2\sqrt{x} - \frac{1}{\sqrt{x}} \right) dx$ i $\int \frac{1}{\sqrt[3]{x}} dx$.

Solució. Escrivint les arrels com a potències, $2x^{1/2} - x^{-1/2}$ i $x^{-1/3}$:

$$\int \left(2x^{1/2} - x^{-1/2} \right) dx = 2 \frac{x^{3/2}}{3/2} - \frac{x^{1/2}}{1/2} + k = \frac{4}{3} x\sqrt{x} - 2\sqrt{x} + k, \quad \int x^{-1/3} dx = \frac{x^{2/3}}{2/3} + k = \frac{3}{2} \sqrt[3]{x^2} + k.$$